Kinematics, Workspace Estimation and Trajectory Planning of Six Robotic Manipulators Using ROS 2 and Gazebo

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By

Vemula Kusumanjali

Registration Number: 231801130015

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Under the guidance of

Dr. N.V.S SHANKAR

Associate Professor

Centurion University of Technology and Management

School of Engineering and Technology

Vizianagaram, Andhra Pradesh, India

Academic Year: 2026–2027

BONAFIDE CERTIFICATE

Certified that this project report **Kinematics, Workspace Estimation and Trajectory Planning of Six Robotic Manipulators Using ROS 2 and Gazebo** is the bonafide work of **"Vemula Kusumanjali (231801130015)"** who carried out the project work under my supervision. This is to further certify to the best of my knowledge that this project has not been carried out earlier in this institute and the university.

SIGNATURE
Dr. N.V.S SHANKAR
Associate Professor

Certified that the above-mentioned project has been duly carried out as per the norms of the college and statutes of the university.

SIGNATURE
K. JOGI NAIDU
HOD (ECE)

SIGNATURE

Dr. SATYABRATHA LENKA

DEAN, SoET

HEAD OF THE DEPARTMENT / DEAN OF THE SCHOOL

Department of Electronics and Communication Engineering
DEPARTMENT SEAL

DECLARATION

I, Vemula Kusumanjali, declare that this report entitled **Kinematics, Workspace Estimation and Trajectory Planning of Six Robotic Manipulators Using ROS 2 and Gazebo** documents the educational robot models, software integration, numerical checks and recorded simulation results described in the report.

The report distinguishes calculated kinematic outputs from measured simulator feedback. It does not represent the educational geometries as calibrated commercial robots, and it does not present the proposed UR5e adaptive-workspace extension as completed experimental work. External software, technical references and supporting methods are acknowledged where used.

I will verify the institutional details and evidence accompanying this report before signing and submitting it for academic assessment.

ACKNOWLEDGEMENT

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I also thank the faculty of the Department of Electronics and Communication Engineering and those who reviewed the demonstrations and helped refine the presentation of the project.

The development relied on the open-source ROS 2, RViz, Gazebo, Bullet, NumPy and SciPy ecosystems. Their documentation and software made it possible to connect mathematical robot models to interactive demonstrations and reproducible simulation analysis.

Finally, I thank my family and friends for their support throughout the implementation, testing and documentation of this project.

Vemula Kusumanjali

ABSTRACT

This project develops a simulation-based laboratory for six robotic manipulators: SCARA, Stanford-type, PUMA-type, SCORBOT-type, Delta and Stewart platform. The objective is to connect forward and inverse kinematics, live parameter tables, sampled workspace estimation and smooth waypoint motion within a common ROS 2 interface. Four serial models use standard Denavit–Hartenberg transformations. Delta and Stewart mechanisms use geometry-based closed-loop constraints. The models are educational and are not calibrated replicas of commercial manipulators.

A Python graphical interface displays actuator values and tool poses, solves inverse-kinematics targets, samples workspaces and executes quintic actuator-space trajectories. RViz displays the calculated geometry. Dynamic verification is handled separately: serial mechanisms use articulated Gazebo models and joint controllers, while parallel mechanisms use a custom Gazebo System plugin containing native Bullet rigid-body dynamics. Recorded commands and simulator measurements are exported for analysis.

Retained numerical checks cover all six models. Four serial recordings excite joints J1 and J2, while two parallel recordings individually excite all three Delta actuators and all six Stewart legs. The parallel recordings contain 2,909 and 4,699 samples respectively. Maximum recorded constraint gaps are approximately 5.335 micrometres for Delta and 0.1908 micrometres for Stewart. All declared parallel screening gates pass for the prescribed small-motion trials. Serial results include transient tracking errors and nonzero gravity-related offsets, demonstrating why visual motion alone is insufficient to assess simulated control performance.

The principal outcome is an integrated educational platform with reproducible evidence and explicit validation boundaries. Collision-aware industrial task feasibility, hardware validation and adaptive UR5e workspace estimation remain future work.

Keywords: ROS 2; Gazebo; robot kinematics; serial manipulators; parallel manipulators; workspace; trajectory planning; simulation verification.

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CHAPTER 1

INTRODUCTION

1.1 Background and motivation

Robot manipulators connect mechanical geometry, coordinate transformations and control. A simulation laboratory lets learners change joint values, inspect the resulting tool pose and compare calculated motion with measured simulator feedback without physical hardware. ROS 2 provides communication between components [1]; RViz displays geometry and Gazebo supports separate dynamic demonstrations.

Serial manipulators form an open chain: SCARA, Stanford-type, PUMA-type and SCORBOT-type illustrate different joint arrangements. Parallel manipulators connect a platform through multiple branches: Delta and Stewart require simultaneous geometric closure. Each mechanism therefore uses an appropriate model rather than a forced common open-chain representation.

1.2 Objectives

1. Implement six educational models with explicit geometry, limits and units.
2. Display actuator values, model parameters, tool position and orientation.
3. Provide FK, IK, sampled workspaces and smooth waypoint motion.
4. Integrate ROS 2/RViz with separate Gazebo physics demonstrations.
5. Evaluate retained command/feedback data, closure and pose consistency.

1.3 Contribution and scope

The contribution is an integrated educational workflow and reproducible verification package. Established kinematic methods are used; no new general IK algorithm or superiority over other robotics tools is claimed. The recorded trials support only their tested configurations and motions. Industrial accuracy, payload capability, collision-free operation and physical hardware validation are outside the completed scope.

CHAPTER 2

LITERATURE SURVEY

2.1 Robotics foundations

Lynch and Park present rigid-body transformations, kinematics and motion planning in a common mathematical framework [2]. These concepts provide the basis for interpreting the pose of a serial end effector and for formulating closed-loop geometric constraints. This project applies established kinematic methods to educational robot geometries; the application and verification are the focus rather than a claim to new kinematics.

2.2 ROS 2 and visualisation

Macenski and co-authors describe the architecture and use of ROS 2 across robotics applications [1]. Its communication model allows independent components to exchange state information. The laboratory uses this separation to keep the user interface, visualisation and recorded-feedback monitoring distinct. RViz is used to display robot markers and transforms [3]; it is not treated as a rigid-body physics solver.

2.3 Simulation and numerical solution tools

Gazebo System plugins expose simulation lifecycle functions and support custom behaviour within the simulator [4]. This enables the parallel add-on to integrate its dynamics with Gazebo visual entities. Bullet supplies the underlying rigid-body and constraint machinery [5]. SciPy provides bounded nonlinear least-squares optimisation [6], used for numerical inverse kinematics and local closure reconstruction where appropriate.

2.4 Workspace estimation and research extensions

Gaussian-growth workspace sampling was proposed by Peidró and colleagues to improve poorly sampled workspace regions [7]. Consequently, random point clouds and adaptive sampling alone should not be described as novel research contributions. The completed laboratory uses sampled workspaces for geometric interpretation. A separate proposal considers orientation, collision, approach clearance and manipulability for a UR5e assembly scenario; that proposal is discussed only as future work.

CHAPTER 3

PROBLEM STATEMENT AND REQUIREMENTS

3.1 Problem statement

Develop a software laboratory for SCARA, Stanford, PUMA, SCORBOT, Delta and Stewart manipulators in which users can enter actuator coordinates or target poses, inspect the relevant model parameters, estimate reachable position regions and move through a sequence of points. The application must make the relationship between geometry and motion visible and provide separate evidence for simulated dynamic behaviour.

3.2 Functional requirements

Table 3.1: Functional requirements and implemented outputs

ID	Requirement	Output
F1	Select any of six models	Model-specific controls and geometry
F2	Apply actuator coordinates	Updated tool pose and rendered configuration
F3	Inspect robot parameters	DH table for serial chains; geometry tables for parallel models
F4	Solve a target pose	Accepted solution or solver failure message
F5	Estimate position workspace	Sampled point cloud within model limits
F6	Execute waypoints	Timed quintic actuator-space movement and path display
F7	Inspect simulated response	Command and measurement feedback from physics trials
F8	Retain numerical evidence	CSV recordings, summaries, plots and validation reports

3.3 Operating environment and acceptance

The user installation uses Ubuntu 24.04 under Windows WSL 2, ROS 2 Jazzy and Gazebo. The recorded parallel inventory identifies Gazebo Sim 8.15.0, Bullet 3.24, GCC 13.3 and CMake 3.28.3. Python, NumPy, SciPy and Qt support the interface and calculations. Separate numerical checks do not constitute a repeat of the user's simulator trials.

Acceptance is based on observed functionality and retained evidence. Serial recordings excite J1/J2; parallel trials use individual small actuator excursions around home. Full-workspace dynamics and physical performance are not established.

CHAPTER 4

SYSTEM ARCHITECTURE AND METHODOLOGY

4.1 Three execution paths

The architecture separates three paths. First, the kinematic path calculates robot geometry and sends markers to RViz; an optional Gazebo mirror displays the same calculated geometry. Second, the serial physics path creates articulated SDF models and uses Gazebo joint controllers. Third, the parallel physics path uses a custom Gazebo System plugin that owns a native Bullet dynamics world. Only the latter two paths provide recorded dynamic measurements.

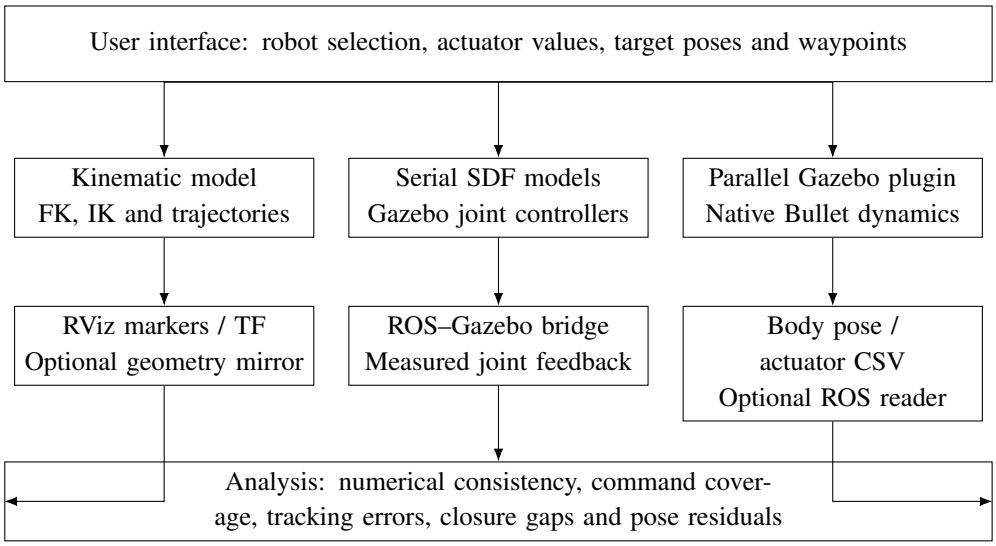


Figure 4.1: Software architecture. The visualisation and dynamic-feedback paths remain distinguishable.

4.2 Software components

Table 4.1: Main implementation components

Component	Responsibility
`kinematics.py`	Robot definitions, transformations, IK and closure functions
`app.py`	Qt interaction, trajectory updates, tables and ROS outputs
`gazebo_mirror.py`	Optional display of calculated geometry
`serial_sdf.py`	Articulated serial model generation and coordinate mapping
`physics_lab.py`	Serial simulation launch and command/feedback interface
`parallel_bullet_lab_v1`	Parallel dynamics core, Gazebo plugin and trial scripts
Analysis scripts	CSV statistics, pose consistency checks and plots

4.3 ROS interfaces and frames

The consolidated interface publishes marker arrays on `/robot_lab/markers`, actuator information on `/robot_lab/joint_states` and a transform from `world` to `lab_tool`. Its marker-based display supports both serial and closed-loop mechanisms. The earlier SCARA demonstration used a robot-state-publisher/TF workflow; it should not be confused with the consolidated marker implementation.

Parallel feedback can be republished as joint states and a tool pose under `/parallel_lab/<robot>/`. These messages originate from the recorded simulator data. File-based republication supports monitoring and does not implement a hard-real-time controller.

4.4 Working procedure

1. Select a robot and load its geometry, actuator limits and home configuration.
2. Apply actuator values and compute FK, or solve IK for a requested target pose.
3. Update parameter tables and visual geometry from the accepted configuration.
4. Add waypoints and generate timed actuator-space segments.
5. Sample allowed configurations or poses to estimate the position workspace.
6. Run the separate physics trial when dynamic feedback is required.
7. Compare commands and measurements, retain CSV evidence and report the limits of each test.

CHAPTER 5

ROBOT MODELLING AND KINEMATICS

5.1 Model families and units

Table 5.1: Implemented educational manipulator structures

Model	Actuated structure	Pose capability illustrated
SCARA	RRPR, 4	Translation and planar tool yaw
Stanford-type	RRPRRR, 6	Serial chain with a prismatic extension
PUMA-type	6R	Six-axis articulated serial motion
SCORBOT-type	5R	Five-axis arm with restricted pose capability
Delta	3R inputs	Translation with fixed platform orientation
Stewart	6P inputs	Platform translation and orientation

Distances are stored in metres and rotations in radians. Degree-valued interface controls are converted at the boundary. The model names indicate educational structures. Their geometry and gains are not manufacturer specifications.

5.2 Standard DH formulation

For a serial link, the standard DH transform is

$$A_i = R_z(\theta_i)T_z(d_i)T_x(a_i)R_x(\alpha_i), \qquad {}^0T_n = \prod_{i=1}^{n} A_i. \tag{5.1}$$

The corresponding homogeneous matrix is

$$A_i = \begin{bmatrix} c_\theta & -s_\theta c_\alpha & s_\theta s_\alpha & a_i c_\theta \\ s_\theta & c_\theta c_\alpha & -c_\theta s_\alpha & a_i s_\theta \\ 0 & s_\alpha & c_\alpha & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{5.2}$$

Here c and s denote cosine and sine. A revolute variable changes θ_i; a prismatic variable changes d_i with the model's specified sign. Multiplication order matters: rotations and translations generally do not commute.

5.3 SCARA model and analytical checks

The arm lengths are 0.30 m and 0.25 m. The consolidated model gives

$$x = 0.30\cos q_1 + 0.25\cos(q_1+q_2), \tag{5.3}$$

$$y = 0.30\sin q_1 + 0.25\sin(q_1+q_2), \tag{5.4}$$

$$z = 0.235 - q_3, \qquad \psi = q_1+q_2+q_4. \tag{5.5}$$

The negative sign in z means that increasing the third coordinate lowers the tool. During the earlier TF demonstration, changing q_3 from 0.050 to 0.100 m changed tool height from 0.185 to 0.135 m. This checks the implemented coordinate convention.

For a target (x,y,z,ψ), define

$$D = \frac{x^2+y^2-L_1^2-L_2^2}{2L_1L_2}, \qquad q_2 = \operatorname{atan2}(\pm\sqrt{1-D^2}, D). \tag{5.6}$$

Then

$$q_1 = \operatorname{atan2}(y,x) - \operatorname{atan2}(L_2\sin q_2, L_1+L_2\cos q_2), \tag{5.7}$$

$$q_3 = 0.235 - z, \qquad q_4 = \psi - q_1 - q_2. \tag{5.8}$$

The two signs give candidate elbow configurations. Candidates must also satisfy actuator limits. The early test target $(0.30, 0.20, 0.15)$ m with zero yaw produced approximately $(-9.59^\circ, 98.63^\circ, 0.085\,\text{m}, -89.04^\circ)$ and $(76.97^\circ, -98.63^\circ, 0.085\,\text{m}, 21.66^\circ)$. These are two configurations for the same tool pose.

A target with $x = 1$ m and $y = 0.20$ m is outside the two-arm geometric reach because its planar radius exceeds $L_1 + L_2 = 0.55$ m. Rejection of this target checks one clear infeasible case; it is not an exhaustive IK validation.

5.4 Stanford, PUMA and SCORBOT models

The Stanford-type model has a third prismatic coordinate with range 0.10–0.55 m. The PUMA-type model uses six revolute coordinates. The SCORBOT-type model uses five revolute coordinates and cannot generally achieve an arbitrary six-dimensional target pose. Their DH parameters are listed in the accompanying `model_parameters.json` and `model_tables.tex` files so the model is reproducible.

For a desired position $\mathbf{p}_d$ and orientation R_d, the consolidated serial solver minimises the bounded residual

$$\mathbf{r}(\mathbf{q}) = \begin{bmatrix} \mathbf{p}(\mathbf{q}) - \mathbf{p}_d \\ \operatorname{Log}(R_d^T R(\mathbf{q}))^\vee \end{bmatrix}. \tag{5.9}$$

The rotation-vector term describes orientation disagreement without directly subtracting Euler angles. The implementation uses the current configuration, home and six deterministic random seeds, with up to 250 function evaluations per solve. Acceptance uses $\|\mathbf{r}\|_2 < 10^{-5}$. Position and orientation residuals are unweighted metres and radians; this threshold is a numerical acceptance rule rather than a dimensionally homogeneous physical error.

Failure from these seeds means that the solver did not find a valid solution within its limits. It does not generally prove that no solution exists. Likewise, passing near-home tests does not prove correctness

at every singular or boundary configuration.

5.5 Delta closure model

The educational Delta has base radius 0.28 m, platform radius 0.075 m, upper-arm length 0.20 m, lower-arm length 0.48 m and base height 0.85 m. For upper-arm endpoint $\mathbf{e}_i(q_i)$ and platform attachment offset $\mathbf{a}_i$, each lower branch imposes

$$\|\mathbf{p} + \mathbf{a}_i - \mathbf{e}_i(q_i)\|_2 = 0.48, \qquad i = 1,2,3. \tag{5.10}$$

Forward kinematics intersects the three equal-radius spheres and selects the lower assembly branch. Inverse kinematics obtains arm-angle candidates, checks limits and verifies consistency with the same branch. Platform orientation is fixed in this geometric model. A valid length equation alone is insufficient if a solution belongs to a different assembly branch.

5.6 Stewart platform closure model

The Stewart platform uses base radius 0.32 m and platform radius 0.20 m. Base anchor angles are $[10,110,130,230,250,350]^\circ$; platform angles are $[50,70,170,190,290,310]^\circ$. For platform pose $(\mathbf{p},R)$, leg lengths follow

$$\ell_i = \|\mathbf{p} + R\mathbf{a}_i - \mathbf{b}_i\|_2, \qquad i = 1,\ldots,6. \tag{5.11}$$

The home pose is $(0,0,0.48)$ m with zero orientation. Leg lengths are limited to 0.30–0.75 m. Inverse kinematics directly evaluates the lengths. Forward kinematics numerically reconstructs a pose near the home or previous pose. It is a local closure solution, not a search over every possible assembly mode.

CHAPTER 6

WORKSPACE ESTIMATION AND TRAJECTORY PLANNING

6.1 Sampled position workspaces

The geometric position workspace is the set of tool positions obtainable from allowable configurations. Finite sampling approximates this set. Retained exports contain 1,200 accepted points for each serial model and Delta, and 1,148 for Stewart. Serial models allow orientation to vary; the Stewart sampling uses a zero-orientation slice. The earlier SCARA demonstration used a denser 12,000-point display. These are different display/export settings, not contradictory measurements.

A sampled point cloud does not certify exact workspace volume. Uniform joint samples are generally nonuniform in Cartesian position, and empty regions can result from insufficient sampling. Collision checks are not included in these workspace plots. A position in the cloud is also not a guarantee that every desired orientation is achievable there.

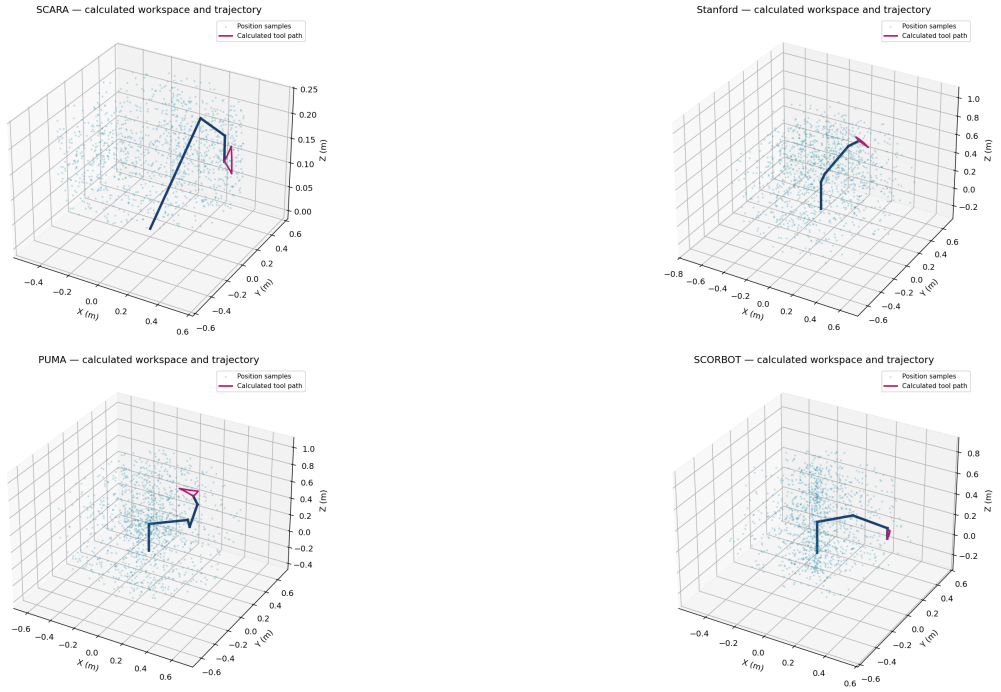


Figure 6.1: Calculated position-workspace samples for the four serial educational models. These plots are model outputs, not physical measurements.

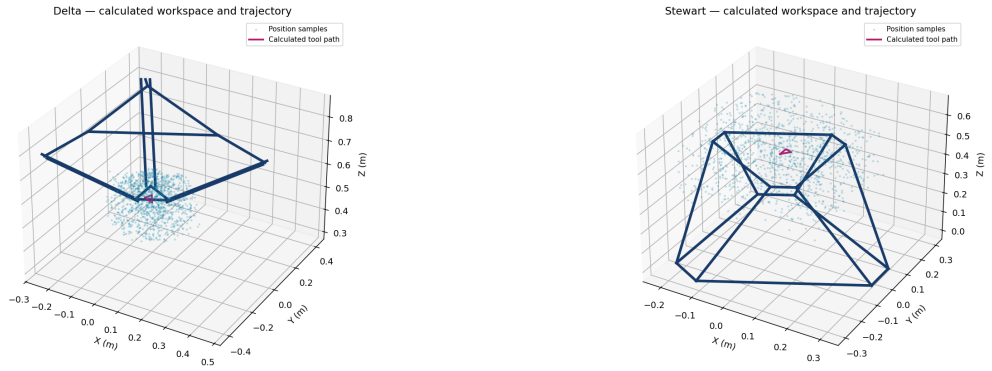


Figure 6.2: Calculated parallel-model workspace samples. Delta has fixed platform orientation; Stewart illustrates a zero-orientation position slice.

6.2 Quintic time scaling

For actuator vectors $\mathbf{q}_a$ and $\mathbf{q}_b$, a segment of duration T uses

$$\mathbf{q}(t) = \mathbf{q}_a + (\mathbf{q}_b - \mathbf{q}_a)s(u), \quad u = t/T, \quad s(u) = 10u^3 - 15u^4 + 6u^5. \tag{6.1}$$

This polynomial has zero velocity and acceleration at each end. All coordinates share the same normalised time, so they arrive together. The interpolation is in actuator space; a visually smooth end-effector path need not be a straight Cartesian line.

The maximum normalised speed is 1.875 and the maximum absolute normalised acceleration is approximately 5.774. Thus the duration is selected to satisfy

$$T \geq \max_i \left(T_{\text{requested}}, \frac{1.875|\Delta q_i|}{v_{i,\max}}, \sqrt{\frac{5.774|\Delta q_i|}{a_{i,\max}}} \right). \tag{6.2}$$

The educational limits are 0.8 rad/s and 1.6 rad/s^2 for revolute coordinates and 0.12 m/s and 0.25 m/s^2 for prismatic coordinates. These are implementation choices rather than actuator ratings.

6.3 Waypoint execution and path checks

The interface can load demonstration waypoints, add target waypoints, clear a sequence, run the sequence and export motion values. Each segment stops at its endpoint. Parallel closure is sampled along the proposed motion because valid endpoints alone do not establish a valid intermediate path. Discrete checking still cannot guarantee continuous feasibility or collision avoidance.

CHAPTER 7

GAZEBO DYNAMIC IMPLEMENTATION

7.1 Serial articulated models

The serial SDF generator places links at the model's home configuration. Simulator coordinates are offset from model coordinates:

$$q_{\mathrm{gz},i} = q_i - q_{\mathrm{home},i}. \tag{7.1}$$

Limits are shifted consistently, and feedback is converted back before comparison. Failing to apply this mapping would create apparent tracking errors even if the simulated joint were at its intended position.

The base is fixed. Nonzero-length links use cylindrical inertias and short links use spherical inertias. Gravity acts along negative z at 9.81 m/s^2, the nominal simulation step is 1 ms and serial self-collision is disabled. Joint controllers use $(K_P, K_I, K_D) = (120, 5, 8)$ for revolute axes and $(600, 5, 30)$ for prismatic axes, with effort limits of ± 100 in the corresponding units.

ROS command topics follow `/physics_lab/<robot>/cmd/joint_<i>`. Serial tool positions in the feedback analysis are FK reconstructions from measured joints. They are not independent Cartesian sensor measurements.

7.2 Native Bullet parallel plugin

The parallel add-on owns a Bullet dynamics world inside a Gazebo System plugin. The standard Gazebo Physics system is omitted from these particular worlds so that two solvers do not update the same mechanism. The plugin steps the bodies and constraints, then copies the resulting body transforms to Gazebo entities for rendering.

This path is different from imposing FK poses in the optional geometry mirror. Mass, inertia, gravity, constraints and actuator forces/torques determine the simulated response. The Bullet solver uses 120 iterations and internal steps no larger than 1 ms. Contacts are disabled for the retained trials.

Table 7.1: Parallel dynamics model and controller settings

Parameter	Delta	Stewart
Actuation	3 hinge torques	6 axial leg forces
Moving platform mass	0.30 kg	1.00 kg
Branch body masses	Upper arm 0.20 kg; lower rod 0.06 kg	Each half-leg 0.15 kg
PID gains (K_P, K_I, K_D)	$(50, 15, 2.5)$	$(2500, 600, 60)$
Saturation	± 15 N m	± 100 N
Integral-state clamp	$[-0.2, 0.2]$	$[-0.2, 0.2]$
Contact model in trials	Disabled	Disabled

7.3 Mechanisms and feedback

Delta uses three driven upper arms and six lower rods connected in paired branches. Stewart uses telescoping legs with point constraints at the base and platform and slider constraints between half-legs. Slider rotation is locked. This is an educational SPS-like realisation; universal-joint bearing-angle limits and calibrated actuator details are not represented.

The control law has proportional and integral position feedback with measured-velocity damping:

$$u_i = \mathrm{sat}[K_P(q_{d,i} - q_i) + K_I I_i - K_D \dot{q}_i]\,. \tag{7.2}$$

Delta receives hinge torque; Stewart receives equal-and-opposite axial forces on the two halves of each leg. Gains and masses are illustrative and require separate evaluation for other configurations or loads.

Actuator measurements are reconstructed from simulated body geometry. The platform pose is taken directly from the simulated platform body. Feedback is recorded at approximately 50 Hz with simulator timestamps; it includes commands, measurements, pose and the recorded constraint-gap diagnostic. The optional Python reader republishes these samples to ROS.

CHAPTER 8

EXPERIMENTAL PROCEDURE AND VERIFICATION

8.1 Evidence categories

Three evidence categories are retained: numerical software checks, user-recorded visualisations and dynamic CSV recordings. They answer different questions. A screenshot establishes what was displayed; a numerical FK–IK round trip checks an implementation relationship; a simulator recording permits assessment of the simulated response under specified conditions.

The numerical validation report explicitly states that its script did not execute ROS or Gazebo. User-side physics recordings are the separate evidence for runtime trials. Combining the evidence requires preserving this distinction.

8.2 Numerical validation

The retained numerical suite checked each of the six models at home and at 12 near-home FK–IK round-trip cases. It also checked selected path closure, actuator timing and limits, generated SDF XML and rejection of a distant target. A separate serial-model suite checked articulated topology, positive inertia and agreement of the joint-coordinate mapping with DH at 80 random configurations per serial model.

These tests passed in the retained report. They establish consistency of the implemented educational models across those cases. They do not independently validate the geometry of a commercial robot or demonstrate global solver completeness.

8.3 Serial trial protocol

The four retained serial recordings vary J1 through 0.15 rad and J2 through 0.12 rad, returning to home. Other joint commands remain fixed. Different run lengths and holding intervals affect aggregate RMSE, so the four records are descriptive case studies rather than a controlled performance ranking.

The SCARA recording contains 5,131 samples including 63 duplicate simulation timestamps. All recorded rows are retained for sample-weighted statistics. The resulting RMSE is therefore not a time-integrated RMS over uniquely weighted timestamps.

8.4 Parallel trial protocol

After an initial 5 s hold, each parallel actuator is exercised individually for 12 s: 3 s outward motion, 3 s hold, 3 s return and 3 s at home. Excursions are 0.04 rad per Delta actuator and 0.004 m per Stewart leg. A final 10 s hold gives planned trial durations of 51 s and 87 s. Gazebo recordings continue to approximately 60 s and 96 s, respectively.

This protocol confirms individual actuator excitation near home. It does not exercise every combination of simultaneous actuator motion. Command spans are checked directly from the CSV data rather than

inferred from animation.

8.5 Error metrics and screening gates

For sample k and actuator i, define $e_i[k] = q_{m,i}[k] - q_{d,i}[k]$. The reported quantities are

$$E_i^{\text{peak}} = \max_k |e_i[k]|, \tag{8.1}$$

$$E_i^{\text{final}} = |e_i[N]|, \tag{8.2}$$

$$E_i^{\text{tail}} = \max_{t_k \geq t_{\text{end}} - 5} |e_i[k]|, \tag{8.3}$$

$$\text{RMSE}_i = \sqrt{\frac{1}{N} \sum_{k=1}^{N} e_i[k]^2}. \tag{8.4}$$

Final-sample error and maximum error in the final five seconds are distinct. A small final value does not erase transient error earlier in the trial.

Every fifth parallel CSV sample is checked against independently implemented FK evaluated at the measured actuator coordinates. The residuals are

$$e_p = \|\mathbf{p}_{\text{body}} - \mathbf{p}_{\text{FK}}\|_2, \qquad e_R = \|\operatorname{Log}(R_{\text{FK}}^T R_{\text{body}})^\vee\|_2. \tag{8.5}$$

The computational paths are separate, but they share nominal geometry; the comparison is not a physical measurement.

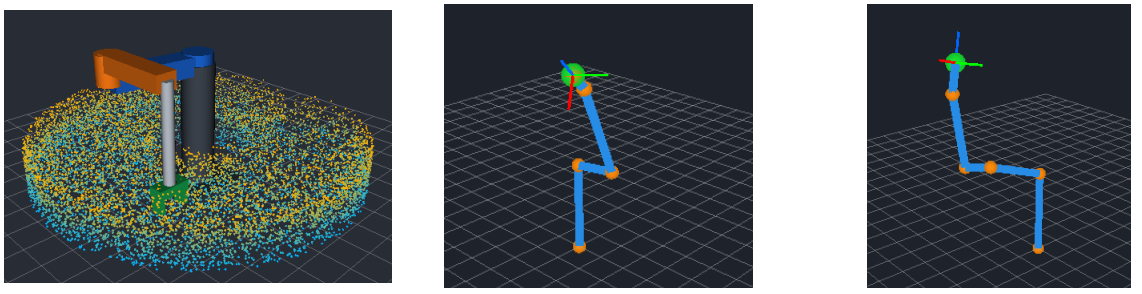
Parallel screening requires finite values, increasing timestamps, full trials, excitation of each actuator, measured limits within the checker's tolerance, recorded gap below 1 mm, sampled FK position residual below 1 mm and rotation residual below 0.01 rad. Tail tracking limits are 0.01 rad for Delta and 1 mm for Stewart. These are developer-selected investigation gates, not industrial acceptance specifications.

CHAPTER 9

RESULTS AND DISCUSSION

9.1 Six-model visual demonstration

Figure 9.1 presents the six manipulator families from user-recorded views. The SCARA panel comes from the earlier workspace demonstration; Stanford, PUMA and SCORBOT are shown in the consolidated RViz application; Delta and Stewart are shown in Gazebo. Different scales and configurations are used for readability.



(a) SCARA: RViz workspace

(b) Stanford-type: RViz

(c) PUMA-type: RViz

(d) SCORBOT-type: RViz

(e) Delta: Gazebo

(f) Stewart: Gazebo

Figure 9.1: Six educational manipulators in recorded displays. The images establish rendering and are not accuracy measurements.

9.2 Serial recorded results

Table 9.1: Serial recordings and excited-joint RMSE

Model	Samples	Time range (s)	J1 RMSE (°)	J2 RMSE (°)
SCARA	5131	0.068–255.272	0.09166	0.08957
Stanford	4683	0.068–223.278	0.28340	0.26880
PUMA	3258	0.068–157.862	0.31114	0.65648
SCORBOT	5169	0.986–249.798	0.16837	0.37221

Table 9.2: Peak and final absolute serial tracking errors for excited joints

Model	J1 peak (°)	J2 peak (°)	J1 final (°)	J2 final (°)
SCARA	1.42381	1.22554	0.0000355	0.004048
Stanford	2.45232	2.19068	0.000146	0.000206
PUMA	2.39784	2.71810	0.000566	0.005526
SCORBOT	2.02604	2.53900	0.0000115	0.003381

The traces show transient deviations when commands change, followed by smaller errors during holding. All four records exhibit the intended J1/J2 command spans. Different geometries, gains, units and hold durations prevent these statistics from supporting a general ranking of the robots.

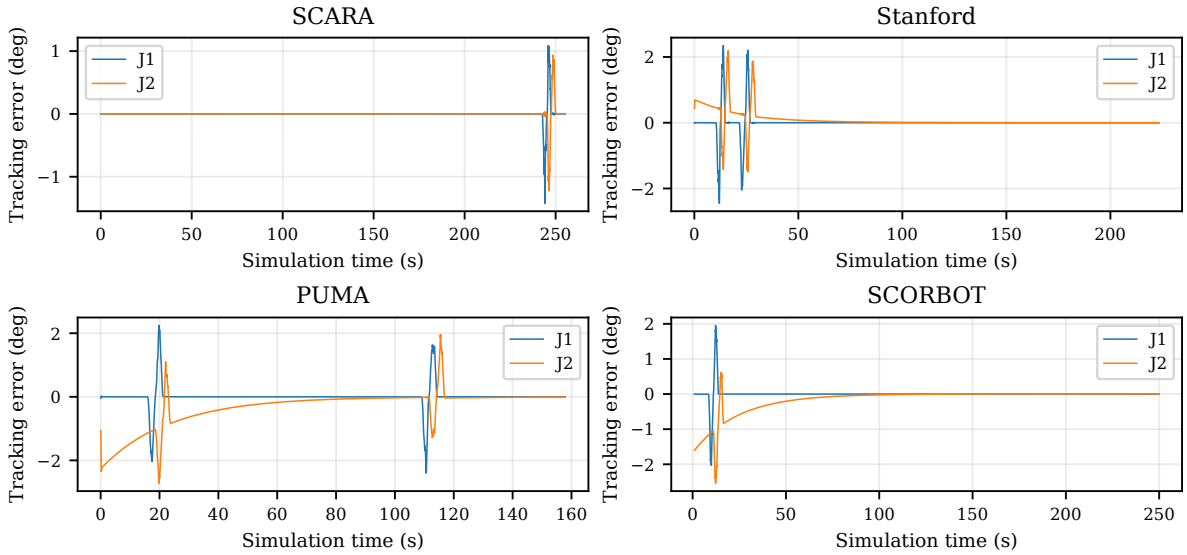


Figure 9.2: Measured-minus-commanded J1 and J2 angles in the four serial recordings.

Held coordinates also matter. Final absolute prismatic J3 errors are approximately 0.916 mm for SCARA and 2.107 mm for Stanford. A stationary command therefore does not imply zero position error in a gravity-loaded simulation. No universal pass/fail threshold for serial tracking was declared for these original recordings.

The recordings predate later launcher timing changes. They document the runtime that produced them and should not be interpreted as regression coverage for every subsequent code change.

9.3 Parallel recorded results

Table 9.3: Parallel Gazebo trial statistics; maxima are across actuators where applicable

Metric	Delta	Stewart
Recorded samples	2909	4699
Final simulator time (s)	59.998	95.988
Excursion per actuator	0.04 rad	4 mm
Peak absolute actuator error	0.01414 rad	1.3359 mm
Final-five-second maximum error	2.1935×10^{-6} rad	6.557×10^{-4} mm
Maximum recorded constraint gap (μm)	5.335	0.1908
Sampled FK position residual (μm)	5.924	0.2446
Sampled FK orientation residual (rad)	1.110×10^{-4}	5.941×10^{-7}
Declared investigation gates	Passed	Passed

Every Delta input and every Stewart leg has a nonzero command span in its recording. Early transients are larger than the final-five-second errors. The displayed traces are therefore necessary to interpret the small tail values in context.

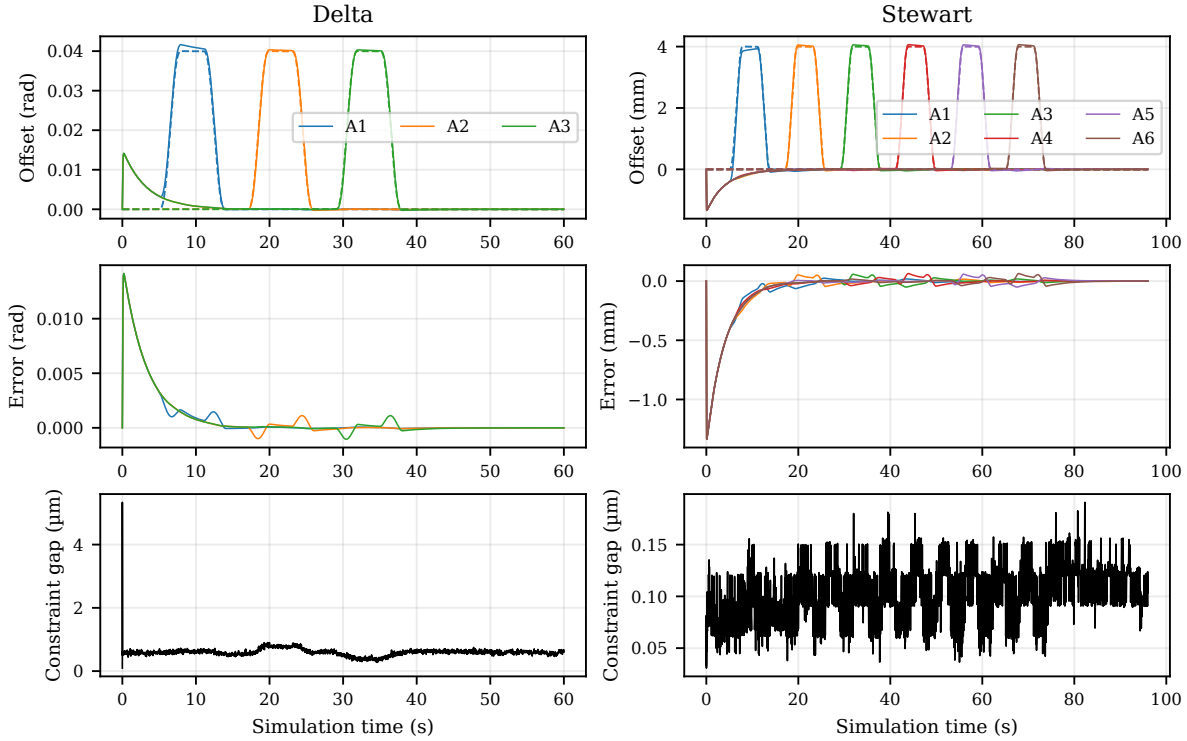


Figure 9.3: Parallel actuator offsets, tracking errors and recorded closure diagnostics. Commands are dashed and measurements are solid in the top row.

The constraint-gap and FK residual values are numerical consistency results in idealised simulations. Their micrometre-scale magnitudes do not establish micrometre physical positioning accuracy. They are also not a basis for claiming Stewart is intrinsically more accurate than Delta, because mechanisms, units, geometry and controller settings differ.

CHAPTER 10

CONCLUSION AND FUTURE WORK

10.1 Conclusion

Six educational manipulator models share a workflow for inspecting parameters, solving target poses, sampling workspaces and executing quintic actuator-space trajectories. Separate dynamic demonstrations expose the difference between commanded and measured motion. Numerical checks and retained recordings support the reported scope: serial trials show transient and holding errors, while Delta and Stewart small-motion trials satisfy the declared closure and tracking gates.

10.2 Limitations and future work

Geometry, inertia and controller gains are illustrative. Friction, backlash, compliance, noise and payload robustness were not identified from hardware. Collision/contact checks are disabled. Serial trials excite only J1/J2, and parallel trials use small excursions around home. Local solvers and finite sampling do not prove global reachability or continuous path feasibility.

Next steps are all-joint serial tests, repeated trials, timestep/gain sensitivity and larger parallel motions with assembly-mode monitoring. A separate UR5e proposal adds task orientation, approach clearance, collision and manipulability to adaptive workspace estimation. It requires new implementation and matched-budget baseline experiments. Adaptive sampling has prior literature [7]; its novelty and performance cannot be inferred from this educational laboratory.

Source code, model parameters and results

The complete implementation, model parameter tables, launch instructions, original feedback CSV files and supporting analysis are available at:

`https://github.com/KUSUMANJALI2027/six-manipulator-ros2-gazebo`

Use `model_parameters.json` and `dh_tables/` for geometry; `evidence/` for recorded simulator feedback; `reports/` for plots and calculated exports; and `documentation/overleaf/` for editable report files. Extended parameter tables are retained in `model_tables.tex` in the Overleaf package. The repository README gives installation and launch instructions.

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